

SMART RECEPTIONIST With SMARTLOCK SYSTEM

BISWAJIT DAS

Presented here is a security system that lets you see a visitor while your main office door is locked. If you are in the middle of a meeting in a conference room and there is a visitor at the door, this system will send a notification to your smartphone with a photo of the visitor as email. If you approve, you either use your mobile or PC to

unlock the front door using a Web browser. Alternatively, if you have a door bell with intercom facility, you can talk to the visitor when he or she rings the door bell.

A typical block diagram of the smart receptionist with a smartlock system is shown in Fig. 1.

Circuit and working

Interfacing circuitry of the smart receptionist and lock system is shown in Fig. 2. Raspberry Pi runs on standard Raspbian Linux distribution with Wi-Fi dongle, GPIO library and software written in Python language. Raspberry Pi GPIO4, GPIO17 and GPIO25 are connected with resistors R3, R2 and R1, respectively, to make logic level low.

Door bell. The door bell signal from CON3 is routed to GPIO17 pin on Raspberry Pi. When a visitor rings the door bell, the status LED glows and GPIO17 becomes high. Amplitude of the door bell signal output depends on the door bell/chime used. You can use a door bell

that gives 3V signal.

Captured signal should not exceed Raspberry Pi's 3.3V limit on GPIO pins. So a 3V zener (ZD1) is used in this circuit. Ground wire of the door bell circuitry should be connected to GND pin of Raspberry Pi and the anode of status LED to GPIO17.

When GPIO17 becomes high, the system captures the photo of the visitor through the webcam connected to Raspberry Pi through USB. Captured photo is sent to your email ID configured in your source program.

Open the attachment in your email and check the photo of the visitor. After your visitor is identified, either open the door directly or talk to him or her through an intercom if it is installed at the door.

Smartlock. When status LED glows to indicate that the door bell is ringing, press Camera tab on your Web browser (Fig. 3). This will make GPIO25 pin of Raspberry Pi high, enabling the webcam to

capture the photo of the visitor. Captured photo is sent to your email ID.

After your visitor is identified, you may open the door by pressing Lock tab on the Web browser. With Lock tab pressed, GPIO4 becomes high. Since relay driver transistor T2 is connected to GPIO4, T2 conducts and energises relay (RL1). This provides a 12V DC supply from CON2 to the solenoid lock connected at CON1, and the door opens.

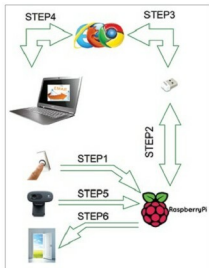


Fig. 1: Smart receptionist with a smartlock system

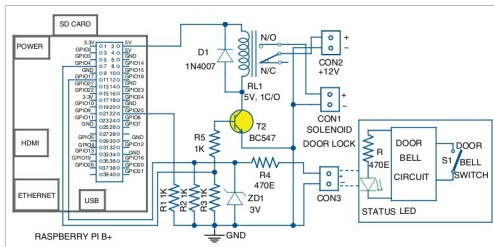


Fig. 2: Circuit diagram of the smart receptionist with a smartlock system